

Homework 4 Message Queue

https://github.com/hhafiya/software_architecture.git, branch micro_mq

Завданням цієї роботи було додати чергу повідомлень для взаємодії facade-service та counter-service, я використовувала **Hazelcast Queue**. Потрібна вона для того, щоб facade-service(producer) не очікував результату від counter-service, а натомість поклав повідомлення у чергу і одразу повернув відповідь клієнту про підтвердження запиту - це суттєво зменшить Response Time. Counter-service(consumer) ж постійно слухає чергу і коли з'являється нове повідомлення, забирає його та обробляє. Для GET запиту ніяких змін не відбулось.

Крім того додався **config-server** - механізм знаходження сервісів. Кожен сервіс при запуску надсилає свою адресу та ім'я в config-server. Config-server зберігає список доступних інстансів для кожного сервісу. І наприклад, коли facade-service потрібно звернутися до logging-service або counter-service, він робить запит до config-server і отримує список доступних адрес. Після цього випадковим чином обирається одна з них. Перевагою такого підходу є те, що немає жорстко прописаних адрес і сервіси шукають одне одного

▼	●	software_architecture	-	-	-	16.03%	1.77GB / 116.7Gi	15.13%	245.23MB / 74.6	281.87KB / 247.2	3, 78, 81, 78, 66, 1	2 seconds ago	■	:
	●	config-server	0d6e4821ddef	software_architecture-c	8001:8000	0.14%	70.81MB / 11.67i	0.59%	9.16MB / 8.6MB	5.55KB / 1.69KB	3	1 minute ago	■	:
	●	hz-node-2-1	3ca241380035	hazelcast/hazelcast.5:	5702:5701	0.74%	325.5MB / 11.67i	2.72%	30.7MB / 193KB	59.4KB / 115KB	78	1 minute ago	■	:
	●	hz-node-1-1	81d52cea081e	hazelcast/hazelcast.5:	5701:5701	9.61%	297.9MB / 11.67i	2.49%	36.2MB / 193KB	66.2KB / 48.5KB	81	1 minute ago	■	:
	●	hz-node-3-1	5918414910dd	hazelcast/hazelcast.5:	5703:5701	1.19%	309.6MB / 11.67i	2.59%	20.5MB / 197KB	52.7KB / 52KB	78	1 minute ago	■	:
	●	mancenter-1	f46680aca13c	hazelcast/managemen	8081:8080	0.36%	432.6MB / 11.67i	3.62%	112MB / 13MB	3.89KB / 126B	66	1 minute ago	■	:
	●	db-1	9100c908c9cf	postgres:15	5432:5432	0.02%	46.32MB / 11.67i	0.39%	32.7MB / 627KB	3.89KB / 126B	6	1 minute ago	■	:
	●	logging-service-2	9ad2333dd49d	software_architecture-l		0.98%	85.38MB / 11.67i	0.71%	0B / 13MB	6.32KB / 5.34KB	5	39 seconds ago	■	:
	●	logging-service-3	c31565f04057	software_architecture-l		1.09%	81.3MB / 11.67G	0.68%	0B / 13MB	6.4KB / 5.34KB	5	39 seconds ago	■	:
	●	logging-service-1	71f39f40ba03	software_architecture-l		0.97%	81.29MB / 11.67i	0.68%	12.3KB / 13MB	6.02KB / 4.73KB	5	39 seconds ago	■	:
	●	facade-service-1	bab6be4f8637	software_architecture-f	8080:8000	0.93%	78.62MB / 11.67i	0.66%	3.96MB / 12.9ME	71.5KB / 14.4KB	4	1 minute ago	■	:
	●	counter-service	f16db324e286	software_architecture-c		0%	0B / 0B	0%	0B / 0B	0B / 0B		2 seconds ago	■	:

динамічно.

**config-server**
 4e277eb522e8  [software_architecture-config-server:latest](#)
[8001:8000](#)

Logs

Inspect

Bind mounts

Exec

Files

Stats

```
INFO: Started server process [10]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
Registered http://counter-service:8000 as counter-service
INFO: 172.19.0.8:60920 - "POST /register?service_name=counter-service&address=http%3A%2F%2Fcounter-service%3A8000 HTTP/1.1" 200 OK
Registered http://logging-service-1:8000 as logging-service
INFO: 172.19.0.12:41636 - "POST /register?service_name=logging-service&address=http%3A%2F%2Flogging-service-1%3A8000 HTTP/1.1" 200 OK
Registered http://logging-service-2:8000 as logging-service
INFO: 172.19.0.10:52430 - "POST /register?service_name=logging-service&address=http%3A%2F%2Flogging-service-2%3A8000 HTTP/1.1" 200 OK
Registered http://logging-service-3:8000 as logging-service
INFO: 172.19.0.11:57036 - "POST /register?service_name=logging-service&address=http%3A%2F%2Flogging-service-3%3A8000 HTTP/1.1" 200 OK
```

Реєстрація сервісів:

```
• (facade-service) ira@MacBook-Air-Ira software_architecture % for i in {1..10}; do
  curl -X POST http://localhost:8080/transaction \
    -H "Content-Type: application/json" \
    -d "{\"user_id\": \"user_${i}\", \"amount\": $i}"
  echo -e "\nSent transaction $i"
done
{"transaction_id":"1776589759460","status":"accepted"}
Sent transaction 1
{"transaction_id":"1776589759606","status":"accepted"}
Sent transaction 2
{"transaction_id":"1776589759632","status":"accepted"}
Sent transaction 3
{"transaction_id":"1776589759663","status":"accepted"}
Sent transaction 4
{"transaction_id":"1776589759681","status":"accepted"}
Sent transaction 5
{"transaction_id":"1776589759697","status":"accepted"}
Sent transaction 6
{"transaction_id":"1776589759712","status":"accepted"}
Sent transaction 7
{"transaction_id":"1776589759730","status":"accepted"}
Sent transaction 8
{"transaction_id":"1776589759745","status":"accepted"}
Sent transaction 9
{"transaction_id":"1776589759761","status":"accepted"}
Sent transaction 10
```

software_architecture-facade-service-1
08137858882c software_architecture-facade-service:latest
8080:8000

Logs	Inspect	Bind mounts	Exec	Files	Stats
INFO: Started server process [12]					
INFO: Waiting for application startup.					
INFO: Application startup complete.					
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)					
INFO: 151.101.128.223:22334 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:48015 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:62045 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:42662 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:19153 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:40096 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:31264 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:18988 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:22507 - "POST /transaction HTTP/1.1" 200 OK					
INFO: 151.101.128.223:55298 - "POST /transaction HTTP/1.1" 200 OK					

Через HTTP POST записати 10 транзакцій *msg1*-*msg10* через **facade-service**.

logging-service-1
8714e77809d4 software_architecture-logging-service-1:latest

Logs	Inspect	Bind mounts	Exec	Files	Stats
INFO: Started server process [10]					
INFO: Waiting for application startup.					
INFO: Application startup complete.					
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)					
LOG: Registered at http://logging-service-1:8000					
LOG: Received transaction 1776589759606					
INFO: 172.19.0.9:55824 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759681					
INFO: 172.19.0.9:55824 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759761					
INFO: 172.19.0.9:55824 - "POST /log HTTP/1.1" 200 OK					

logging-service-2
804d06909544 software_architecture-logging-service-2:latest

Logs	Inspect	Bind mounts	Exec	Files	Stats
INFO: Started server process [10]					
INFO: Waiting for application startup.					
INFO: Application startup complete.					
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)					
LOG: Registered at http://logging-service-2:8000					
LOG: Received transaction 1776589759632					
INFO: 172.19.0.9:56792 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759697					
INFO: 172.19.0.9:56792 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759730					
INFO: 172.19.0.9:56792 - "POST /log HTTP/1.1" 200 OK					

logging-service-3
61745ab8f284 software_architecture-logging-service-3:latest

Logs	Inspect	Bind mounts	Exec	Files	Stats
INFO: Started server process [9]					
INFO: Waiting for application startup.					
INFO: Application startup complete.					
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)					
LOG: Registered at http://logging-service-3:8000					
LOG: Received transaction 1776589759460					
INFO: 172.19.0.9:60208 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759663					
INFO: 172.19.0.9:60208 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759712					
INFO: 172.19.0.9:60208 - "POST /log HTTP/1.1" 200 OK					
LOG: Received transaction 1776589759745					
INFO: 172.19.0.9:60208 - "POST /log HTTP/1.1" 200 OK					

Показати що повідомлення отримують різні екземпляри **logging-service**.

Через HTTP GET з **facade-service** прочитати транзакції, перевірити що значення на рахунках є коректним.

```
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/accounts
{"user_1":1.0,"user_2":2.0,"user_3":3.0,"user_4":4.0,"user_5":5.0,"user_6":6.0,"user_7":7.0,"user_8":8.0,"user_9":9.0,"user_10":10.0}
```

```

● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_1
{"balance":1.0,"transactions":[{"transaction_id":"1776590188297","user_id":"user_1","amount":1.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_2
{"balance":2.0,"transactions":[{"transaction_id":"1776590188409","user_id":"user_2","amount":2.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_3
{"balance":3.0,"transactions":[{"transaction_id":"1776590188428","user_id":"user_3","amount":3.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_4
{"balance":4.0,"transactions":[{"transaction_id":"1776590188457","user_id":"user_4","amount":4.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_5
{"balance":5.0,"transactions":[{"transaction_id":"1776590188474","user_id":"user_5","amount":5.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_6
{"balance":6.0,"transactions":[{"transaction_id":"1776590188491","user_id":"user_6","amount":6.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_7
{"balance":7.0,"transactions":[{"transaction_id":"1776590188508","user_id":"user_7","amount":7.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_8
{"balance":8.0,"transactions":[{"transaction_id":"1776590188525","user_id":"user_8","amount":8.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_9
{"balance":9.0,"transactions":[{"transaction_id":"1776590188542","user_id":"user_9","amount":9.0}]}%
● (facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_10
{"balance":10.0,"transactions":[{"transaction_id":"1776590188559","user_id":"user_10","amount":10.0}]}%

```

Перевірка відмовостійкості:

1. Вимкнути чи зробити недоступним (docker pause) counter-service та перевірити, що запис транзакцій йде без помилок

```
(facade-service) ira@MacBook-Air-Ira software_architecture % docker pause counter-service
counter-service
(facade-service) ira@MacBook-Air-Ira software_architecture %
```

mancenter-1	ebf750691961	hazelcast/managemen	8081.8080	0.37%	420MB / 11.67Gi	3.51%	193KB / 13.1MB	4.14KB / 126B	60	9 minutes ago	
counter-service	25495d854bcf	software_architecture-c		0%	83.47MB / 11.67Gi	0.7%	0B / 13.1MB	152KB / 84.9KB	7	9 minutes ago	
logging-service-2	c1053d004b93	software_architecture-l		1.21%	81.34MB / 11.67Gi	0.68%	0B / 13MB	67.4KB / 54.2KB	6	8 minutes ago	

```

● (facade-service) ira@MacBook-Air-Ira software_architecture % for i in {1..20}; do
  curl -X POST http://localhost:8080/transaction \
    -H "Content-Type: application/json" \
    -d '{"user_id": "\user_$i"/, "amount": "$i"}'
  echo -e "\nSent transaction $i"
done
{"transaction_id":"1776590671032","status":"accepted"}
Sent transaction 11
{"transaction_id":"1776590671078","status":"accepted"}
Sent transaction 12
{"transaction_id":"1776590671114","status":"accepted"}
Sent transaction 13
{"transaction_id":"1776590671130","status":"accepted"}
Sent transaction 14
{"transaction_id":"1776590671145","status":"accepted"}
Sent transaction 15
{"transaction_id":"1776590671161","status":"accepted"}
Sent transaction 16
{"transaction_id":"1776590671183","status":"accepted"}
Sent transaction 17
{"transaction_id":"1776590671199","status":"accepted"}
Sent transaction 18
{"transaction_id":"1776590671213","status":"accepted"}
Sent transaction 19
{"transaction_id":"1776590671228","status":"accepted"}
Sent transaction 20

```

INFO:	151.101.128.223:41883	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:64212	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:50626	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:24132	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:38970	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:43455	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:29782	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:45598	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:62732	- "POST /transaction HTTP/1.1"	200 OK
INFO:	151.101.128.223:46825	- "POST /transaction HTTP/1.1"	200 OK

При цьому, відповідно GET-запити не мають спрацьовувати, мають або повертати null у якості значень на рахунку

```

(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_11
{"balance":null,"transactions":[],"error":""}%
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_18
{"balance":null,"transactions":[],"error":""}%
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_1
{"balance":null,"transactions":[],"error":""}%
(facade-service) ira@MacBook-Air-Ira software_architecture %
counter-service
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/accounts
{"error":"Counter service unavailable","accounts":[]}%

```

```
(facade-service) ira@MacBook-Air-Ira software_architecture % docker unpause counter-service
counter-service
```

software-architecture	-	-	-	9.71%	1.85GB / 128.37C	15.88%	303.6KB / 89.76B	6.46MB / 6.43ME	83, 3, 72, 7, 73, 61	16 minutes ago		
hz-node-1-1	7197f3262573	hazelcast/hazelcast:5.1	5701:5701	0.92%	358.5MB / 11.67G	3%	0B / 655KB	1.54MB / 2.84ME	83	16 minutes ago		
config-server	e212c01f0c55	software-architecture:c	8001:8000	0.24%	68.6MB / 11.67G	0.57%	0B / 8.6MB	26.8KB / 23.8KB	3	16 minutes ago		
hz-node-2-1	c8943b63b2c0	hazelcast/hazelcast:5.1	5702:5701	0.85%	312.3MB / 11.67G	2.61%	0B / 672KB	2.03MB / 1.56ME	72	16 minutes ago		
db-1	c3c14e5e8804	postgres:15	5432:5432	0.02%	21.28MB / 11.67G	0.18%	49.2KB / 799KB	36.3KB / 33.1KB	7	16 minutes ago		
hz-node-3-1	61622fa8238c	hazelcast/hazelcast:5.1	5703:5701	1.14%	304.7MB / 11.67G	2.55%	61.4KB / 705KB	2.04MB / 1.42ME	73	16 minutes ago		
mancenter-1	ebf750691961	hazelcast/managemen	8081:8080	0.41%	420.2MB / 11.67G	3.52%	193KB / 13.3MB	4.22KB / 126B	60	16 minutes ago		
counter-service	25495d854bcf	software-architecture:c		1.29%	83.94MB / 11.67G	0.7%	0B / 13.1MB	192KB / 103KB	11	16 minutes ago		
logging-service-2	c1053d004b93	software-architecture:l		0.95%	81.34MB / 11.67G	0.68%	0B / 13MB	119KB / 95.7KB	6	16 minutes ago		
logging-service-1	a212a7d45496	software-architecture:l		1.34%	81.36MB / 11.67G	0.68%	0B / 13MB	126KB / 98.9KB	6	16 minutes ago		
logging-service-3	09cbb48ca2bd	software-architecture:l		1.27%	81.36MB / 11.67G	0.68%	0B / 13MB	123KB / 98KB	6	16 minutes ago		
facade-service-1	02e243f31116	software-architecture:f	8080:8000	1.28%	85.41MB / 11.67G	0.71%	0B / 13MB	246KB / 173KB	5	16 minutes ago		

```
INFO: 172.19.0.8:60608 - "GET /balance/user_9 HTTP/1.1" 200 OK
INFO: 172.19.0.8:60608 - "GET /balance/user_10 HTTP/1.1" 200 OK
WORKER: Processing transaction for user_11
WORKER: Processing transaction for user_12
WORKER: Processing transaction for user_13
WORKER: Processing transaction for user_14
WORKER: Processing transaction for user_15
WORKER: Processing transaction for user_16
WORKER: Processing transaction for user_17
WORKER: Processing transaction for user_18
WORKER: Processing transaction for user_19
WORKER: Processing transaction for user_20
INFO: 172.19.0.8:52398 - "GET /balance/user_15 HTTP/1.1" 200 OK
```

```
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_11
{"balance":11.0,"transactions":[{"transaction_id":"1776590671032","user_id":"user_11","amount":11.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_12
{"balance":12.0,"transactions":[{"transaction_id":"1776590671078","user_id":"user_12","amount":12.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_13
{"balance":13.0,"transactions":[{"transaction_id":"1776590671114","user_id":"user_13","amount":13.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_14
{"balance":14.0,"transactions":[{"transaction_id":"1776590671130","user_id":"user_14","amount":14.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_15
{"balance":15.0,"transactions":[{"transaction_id":"1776590671145","user_id":"user_15","amount":15.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_16
{"balance":16.0,"transactions":[{"transaction_id":"1776590671161","user_id":"user_16","amount":16.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_17
{"balance":17.0,"transactions":[{"transaction_id":"1776590671183","user_id":"user_17","amount":17.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_18
{"balance":18.0,"transactions":[{"transaction_id":"1776590671199","user_id":"user_18","amount":18.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_19
{"balance":19.0,"transactions":[{"transaction_id":"1776590671213","user_id":"user_19","amount":19.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/user/user_10
{"balance":10.0,"transactions":[{"transaction_id":"1776590188559","user_id":"user_10","amount":10.0}]}
(facade-service) ira@MacBook-Air-Ira software_architecture % curl http://localhost:8080/accounts
{"user_1":1.0,"user_2":2.0,"user_3":3.0,"user_4":4.0,"user_5":5.0,"user_6":6.0,"user_7":7.0,"user_8":8.0,"user_9":9.0,"user_10":10.0,"user_11":11.0,"user_12":12.0,"user_13":13.0,"user_14":14.0,"user_15":15.0,"user_16":16.0,"user_17":17.0,"user_18":18.0,"user_19":19.0,"user_20":20.0}
(facade-service) ira@MacBook-Air-Ira software_architecture %
```

Після відновлення counter-service всі транзакції пройшли успішно і на рахунках користувачів з'явилися коректні суми.